

Week 6 Lab Worksheet

The objective of this Week 5 laboratory session was to explore the integration of local network emulation with public cloud infrastructure. By utilizing **Open vSwitch (OVS)** to bridge a virtual Mininet topology with a physical network interface, we established end-to-end external connectivity (**LO1 & LO4**). Furthermore, we leveraged **GitHub** and a **Platform-as-a-Service (PaaS)** provider (Render.com) to provision and deploy cloud-hosted resources (**LO2**). Finally, by utilizing standard command-line utilities from within the emulated network, we successfully consumed and validated these cloud-delivered services, demonstrating the practical realization of cloud-to-edge connectivity (**LO3**).

Learning Objectives

1. Bridge virtualized network topologies with physical infrastructure to establish external internet connectivity.
2. Provision and deploy cloud-hosted applications utilizing version control and Platform-as-a-Service (PaaS) architectures
3. Consume and validate cloud-delivered services from emulated edge nodes using standard network utilities.
4. Configure virtual network interfaces and DNS resolution to facilitate seamless Layer 3 routing across heterogeneous environments.

1. Set up GitHub

- a. Go to the link and set up an account (<https://github.com/>)
- b. Once you have registered, create a new repository and call it CloudPage

- c. You will see the following landing page

Quick setup — if you've done this kind of thing before

Set up in Desktop or HTTPS SSH `git@github.com:omeshF/CloudPage.git`

Get started by [creating a new file](#) or [uploading an existing file](#). We recommend every repository include a [README](#), [LICENSE](#), and [.gitignore](#).

...or create a new repository on the command line

```
echo "# CloudPage" >> README.md
git init
git add README.md
git commit -m "first commit"
git branch -M main
git remote add origin git@github.com:omeshF/CloudPage.git
git push -u origin main
```

...or push an existing repository from the command line

```
git remote add origin git@github.com:omeshF/CloudPage.git
git branch -M main
git push -u origin main
```

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- d. Click on **creating a new file** and the name of the file would **index.html** and copy in the content below

```
<!DOCTYPE html>
<html>
<head>
<title>Classroom Cloud</title>
</head>
<body>
<h1>Hello from the Real Cloud!</h1>
</body>
</html>
```

- e. Now click Commit changes. You have now created a simple HTML page in your GitHub. We will launch it in a cloud.

2. Launch Mininet

- a. `sudo mn --topo=single,2`

This will create a simple mininet topology

- b. Open a new terminal.™

- a. `sudo ifconfig s1 up`
- b. `sudo ovs-vsctl add-port s1 ens36`
- c. `sudo ifconfig ens36 0`
- d. `sudo dhclient s1`

- c. Assign IP Addresses to Mininet Hosts. Go to mininet and open

- a. `xterm h1`

In the h1 terminal

- i. `ifconfig h1-eth0 0`
- ii. `dhclient h1-eth0`
- iii. `exit`

- b. `xterm h2.`

In the h2 terminal

- i. `ifconfig h2-eth0 0`
- ii. `dhclient h2-eth0`
- iii. `exit`

- c. `xterm h1`

In the h1 terminal

- i. `echo "nameserver 8.8.8.8" | sudo tee /etc/resolv.conf`
- ii. `echo "nameserver 8.8.4.4" | sudo tee -a /etc/resolv.conf`
- iii. `exit`

- d. Test Connectivity in mininet terminal

- i. `h1 ping h2`
- ii. `h2 ping h1`
- iii. `pingall`
- iv. `h1 ping google.com`

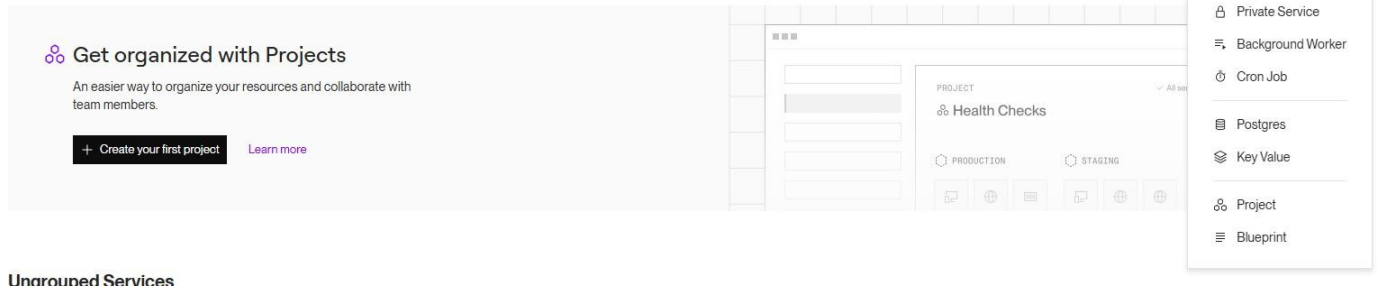
```
mininet-wifi> h1 ping google.com
PING google.com (142.250.140.102) 56(84) bytes of data.
64 bytes from wj-in-f102.1e100.net (142.250.140.102): icmp_seq=1 ttl=128 time=5.38 ms
64 bytes from wj-in-f102.1e100.net (142.250.140.102): icmp_seq=2 ttl=128 time=5.84 ms
64 bytes from wj-in-f102.1e100.net (142.250.140.102): icmp_seq=3 ttl=128 time=6.06 ms
64 bytes from wj-in-f102.1e100.net (142.250.140.102): icmp_seq=4 ttl=128 time=7.58 ms
```

3. Setting up Render.com

Go to render.com (<https://render.com/>) and create an account. After you have registered, on the Dashboard, click **Add New** and choose a **Static Site**

Overview

Projects



Choose the **Public Git Repository** Option and paste the below link <https://github.com/omeshF/simple-cloud> and click **Connect**. On the next page, don't change any settings (you may change the name) can click **Deploy Static Site** at the bottom of the page. You will see the following



Click on the URL (this may change based on the name you gave in the previous step) You will see a page, launched as Cloud Hosted Web Page or a Service.

4. Access using Mininet

Now let's access the cloud web page. Change the URL of the below from the render.com dashboard.

- i. `h1 curl URL`
- ii. `h1 w3m URL`

5. IaaS

Now create another web page on render.com. Use the following GitHub link <https://github.com/omeshF/simple-iaas>

This will create a IaaS launched on render.com. Again the URL may change based on the name you assigned to it. On mininet

- i. `h1 curl -O https://simple-iaas.onrender.com/iaas-explanation.txt`

This will download a text file to your VM., realising an IaaS

6. HelloWorld Web Page (Important)

Now launch the webpage you created in Task 1, on render.com and access it using Mininet. Show the work to your tutor.

7. Extension Task

In this lab, we deployed both a 'Static Site' (Task 3) and an 'IaaS' simulation (Task 5) using Render.com. Contrast these two deployments using the Cloud Shared Responsibility Model. Furthermore, by bridging the Mininet virtual switch directly to the physical network interface (ens36) without encapsulation (like VXLAN) or a firewall, what security risks are introduced to the physical host network if one of the Mininet nodes becomes compromised?